Chapter 3: The Cox regression model — Solutions

STA310 Survival Analysis (Master's in Applied Biostatistics)

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5.1 Fitting a Cox model in R 8

Chapter 3 introduces the Cox proportional hazards model, the central modelling tool for relating survival experience to explanatory variables. These exercises cover the formulation of the model, its fitting via partial likelihood, inference for hazard ratios, model comparison and selection, and the interpretation and use of the fitted model. Model checking (residuals and the proportional hazards assumption) is taken up in Chapter 4.

1 The Cox regression model (Book §3.1–3.2)

1.1 The hazard ratio

Type: Concept & notation

For two treatment groups, the proportional hazards model is $h_N(t) = \psi\, h_S(t)$, where $h_N(t)$ and $h_S(t)$ are the hazards for the new and standard treatments.

(a) What does the quantity ψ represent, and what does the value $\psi = 0.7$ mean in words?
(b) Why is ψ re-expressed as $\exp(\beta)$? What is β called, and what sign of β corresponds to the new treatment being inferior?

(c) For $\beta = -0.3$, compute the hazard ratio and state which treatment is better.

Solution.

(a) ψ is the hazard ratio: the ratio of the hazard of death at time t for an individual on the new treatment to that for an individual on the standard treatment, assumed constant over time. $\psi = 0.7$ means the hazard of death at any time is 30% lower on the new treatment, so the new treatment is better.
(b) The hazard ratio cannot be negative, so it is convenient to write $\psi = \exp(\beta)$; $\beta = \log \psi$ is the logarithm of the hazard ratio. β can be any real value, and a positive β gives $\psi > 1$, i.e. the new treatment is inferior.
(c) $\psi = \exp(-0.3) \approx 0.74$. Since the hazard ratio is below unity, the new treatment is better (lower hazard).

1.2 The baseline hazard and the linear component

Type: Concept & notation

The general Cox model for the hazard of the ith individual is

$$h_i(t) = \exp(\beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_p x_{pi})\, h_0(t).$$

(a) What is $h_0(t)$, and for whom is it the hazard?
(b) Why does the linear component contain no intercept (constant) term β_0?
(c) The quantity $\eta_i = \beta_1 x_{1i} + \cdots + \beta_p x_{pi}$ has two other names. What are they, and how is η_i used to compare two individuals?

Solution.

(a) $h_0(t)$ is the baseline hazard — the hazard for an individual for whom all the explanatory variables equal zero. In a model containing only a treatment indicator, it is the hazard in the reference group.
(b) If a constant β_0 were included, the baseline hazard could simply be rescaled (divided by $\exp(\beta_0)$) and the constant would cancel out, so β_0 is not identifiable and is absorbed into $h_0(t)$.
(c) The linear component is also called the *risk score* or *prognostic index*. The ratio of hazards for two individuals i and k is $\exp(\eta_i - \eta_k)$, so the risk score orders individuals by their hazard — a higher score means a higher hazard.

1.3 Fitting a factor

Type: From research question to model

A study of survival after kidney cancer (hypernephroma) records two factors: age group (< 60, 60–70, > 70) and whether a nephrectomy was performed (no / yes).

(a) How many indicator (dummy) variables are needed to fit age group as a factor, and why? How many degrees of freedom does the factor have?
(b) Taking age group < 60 and "no nephrectomy" as the baseline levels, describe in words the patient to whom the baseline hazard $h_0(t)$ corresponds.

(c) Write the model for the hazard of a patient aged > 70 who had a nephrectomy, in terms of the relevant coefficients.

Solution.

(a) A factor with three levels needs two indicator variables (e.g. $A_2 = 1$ if 60–70, $A_3 = 1$ if > 70); the third level is the baseline. The factor has $3 - 1 = 2$ degrees of freedom. In general an m-level factor needs $m - 1$ parameters.
(b) The baseline hazard corresponds to a patient in the youngest age group (< 60) who did not have a nephrectomy, since all indicator variables are then zero.
(c) With coefficients α_2 (60–70), α_3 (> 70) and ν (nephrectomy), the hazard is $\exp(\alpha_3 + \nu)\, h_0(t)$.

1.4 Proportional hazards and the semi-parametric model

Type: Assumptions

Builds on: Chapters 1–2

(a) In your own words, state the proportional hazards assumption underlying the Cox model, and say what kind of behaviour it rules out.
(b) The baseline hazard $h_0(t)$ is left completely unspecified — no distributional assumption is made about it. Why is the Cox model therefore called semi-parametric?
(c) Why does leaving $h_0(t)$ unspecified still allow the β-coefficients (and hence the hazard ratios) to be estimated?

Solution.

(a) Proportional hazards means the effect of each covariate is a constant multiplicative factor on the hazard at every time t: the hazard for an individual is $\exp(\beta' x)$ times the baseline at all t. It rules out covariate effects that change (or reverse) over time — e.g. a treatment that helps early but harms later.
(b) The "parametric" part is the assumed exponential form of the covariate effects (the βs), while the "non-parametric" part is the unspecified baseline hazard $h_0(t)$. Hence semi-parametric.
(c) In the partial likelihood, $h_0(t)$ cancels from numerator and denominator at each event time, so the βs can be estimated using only the rank order of the survival times, without estimating $h_0(t)$.

2 Fitting the model and making inferences (Book §3.3–3.5)

2.1 The partial likelihood and the risk set

Type: Risk set reasoning

Builds on: Chapter 2

Five patients are followed. Patients 2 and 5 are right-censored, and the three deaths, in order, are patients 3, 1 and 4 at times $t_{(1)} < t_{(2)} < t_{(3)}$. Patient 5 is censored before $t_{(2)}$, and patient 2 between $t_{(2)}$ and $t_{(3)}$. Write $\psi(i) = \exp(\beta' x_i)$.

(a) List the risk set at each of the three death times.

(b) Write the partial likelihood as a product over the three death times.

(c) Censored patients contribute no term to the numerator of the partial likelihood. Where does their information enter, and what does this imply about which feature of the data drives the estimate of β?

Solution.

(a) $R(t_{(1)}) = \{1, 2, 3, 4, 5\}$ (all five); $R(t_{(2)}) = \{1, 2, 4\}$ (patient 3 has died, patient 5 censored); $R(t_{(3)}) = \{4\}$ (patient 1 has died, patient 2 censored).

(b) $L(\beta) = \frac{\psi(3)}{\psi(1)+\psi(2)+\psi(3)+\psi(4)+\psi(5)} \times \frac{\psi(1)}{\psi(1)+\psi(2)+\psi(4)} \times \frac{\psi(4)}{\psi(4)}$.

(c) Censored individuals enter through the denominators (the risk sets) at death times that occur before their censoring time. Because $L(\beta)$ depends only on the ordering of the death times — which determines the risk sets — inference about β depends only on the rank order of the survival times, not their actual values.

2.2 Interpreting Cox regression output

Type: Interpret R output

For the breast cancer data (Example 1.2), a Cox model with staining indicator X (1 = positive, 0 = negative) gives $\hat{\beta} = 0.908$ with $\mathrm{se}(\hat{\beta}) = 0.501$.

(a) Compute the estimated hazard ratio and interpret it in words.

(b) Compute a 95% confidence interval for the log hazard ratio β, and hence for the hazard ratio. Why is the interval obtained by exponentiating the β-limits preferable to $\hat{\psi} \pm 1.96\,\mathrm{se}(\hat{\psi})$?

(c) The null model has $-2\log\hat{L} = 173.968$ and the model with X has 170.096. Carry out the likelihood-ratio test and interpret. (The corresponding log-rank test gave $\chi^2 = 3.515$, $P = 0.061$.)

Solution.

(a) $\hat{\psi} = \exp(0.908) = 2.48$. The hazard of death at any time for a woman with a positively stained tumour is about 2.5 times that for a negatively stained woman; positive staining indicates a poorer prognosis.

(b) A 95% CI for β is $0.908 \pm 1.96 \times 0.501 = (-0.074, 1.890)$. Exponentiating gives $(0.93, 6.62)$ for the hazard ratio. The log hazard ratio is closer to normally distributed than the hazard ratio itself, so exponentiating the β-limits gives a better-behaved interval.

(c) The difference is $173.968 - 170.096 = 3.872$ on 1 degree of freedom, giving $P = 0.049$. There is evidence, just significant at the 5% level, of a difference in hazard between the staining groups — essentially the same conclusion as the log-rank test ($P = 0.061$); the small difference in p-values is immaterial.

2.3 Interpreting a multi-variable model

Type: Interpret R output

For the multiple myeloma data (Example 1.3), a Cox model is fitted with all seven explanatory variables. Of the individual 95% confidence intervals, only those for Bun ($\hat{\beta} = 0.021$, se $= 0.006$)

and Hb ($\hat{\beta} = -0.135$, se $= 0.069$) exclude zero. When Bun is omitted, the estimated coefficient of Hb changes from -0.135 to -0.140, and the other coefficients change too.

(a) Why can we *not* conclude from these individual results alone that only Bun and Hb are needed in the model?
(b) What does the change in the estimated coefficients when Bun is omitted tell us about the estimates?
(c) In a model with several variables, what does it mean to say that a coefficient is "adjusted for" the others?

Solution.

(a) Each individual Wald test tests $\beta_j = 0$ *in the presence of all the other variables*, and the estimates are correlated with one another. A variable's coefficient can change when another is removed, so the relevant subset cannot be identified from a single full model.
(b) It shows the estimates depend on which other variables are in the model (the variables are correlated), so each term's effect must be studied by comparing models, not by inspecting one full fit in isolation.
(c) "Adjusted for" the other variables means the estimated effect of a variable holds the other variables in the model fixed — it is the effect on the hazard controlling for the others.

2.4 Comparing nested models

Type: Model comparison

For the hypernephroma data (Example 3.4), the values of $-2\log\hat{L}$ for five nested models are summarised below. Age group is fitted via A_2, A_3 and nephrectomy via N; the interaction uses the products A_2N, A_3N.

Terms in model	Variables	$-2\log\hat{L}$
none	—	177.667
age group	A_2, A_3	172.172
nephrectomy	N	170.247
age group + nephrectomy	A_2, A_3, N	165.508
1. interaction	A_2, A_3, N, A_2N, A_3N	162.479

(a) Test for an interaction between age group and nephrectomy. What do you conclude?
(b) Does age group add to a model that already contains nephrectomy? Does nephrectomy add to a model that already contains age group?
(c) Which model would you select, and why? What role do the degrees of freedom play in these comparisons?

Solution.

(a) The interaction reduces $-2\log\hat{L}$ by $165.508 - 162.479 = 3.029$ on 2 d.f., $P = 0.220$, which is not significant. There is no evidence of an interaction: the effect of nephrectomy does not depend on age group.

(b) Age group added to nephrectomy: $170.247 - 165.508 = 4.739$ on 2 d.f. ($P = 0.094$), some evidence. Nephrectomy added to age group: $172.172 - 165.508 = 6.664$ on 1 d.f. ($P = 0.010$), significant. So nephrectomy is clearly needed, and age group adds something in the presence of nephrectomy.

(c) Select the model with both main effects and no interaction: the hazard depends on both age group and nephrectomy. The degrees of freedom are the difference in the number of parameters between the two nested models, which is the degrees of freedom of the chi-squared distribution of the difference in $-2\log\hat{L}$.

3 Model selection (Book §3.6–3.8)

3.1 A strategy for model selection

Type: Method selection

(a) State the hierarchic principle, and give an example of a situation where it applies.

(b) Why does the book recommend against relying uncritically on automatic forward selection, backward elimination or stepwise procedures?

(c) Describe the first step of the recommended model-selection strategy, and explain why a relatively large significance level (about 15%) is suggested for guiding inclusion.

Solution.

(a) The hierarchic principle: a higher-order term (e.g. an interaction) should be included only when the corresponding lower-order (main effect) terms are present; otherwise the model is difficult to interpret. For example, an interaction between age group and nephrectomy is only interpretable when both main effects are fitted.

(b) Automatic routines typically return a single "best" subset rather than a set of equally good models; the result depends on the direction (forward, backward, stepwise) and on the stopping rule; and they generally ignore the hierarchic principle.

(c) Step 1 fits each variable alone and compares its $-2\log\hat{L}$ with that of the null model, to find which variables on their own significantly reduce the statistic. A generous level (about 15%) avoids excluding potentially relevant variables too early, since a variable unimportant on its own may become important in the presence of others.

3.2 Variable selection with the lasso

Type: Method selection

(a) What does the lasso do to the estimated coefficients, and how does it carry out variable selection?

(b) Why must the explanatory variables be standardised before the lasso is applied?

(c) How is the optimal value of the tuning parameter λ chosen, and why are standard errors and hazard ratios usually not reported after using the lasso?

Solution.

(a) The lasso shrinks the coefficients towards zero by penalising the sum of their absolute values (the L_1-norm); some are set exactly to zero, so the corresponding variables are excluded. The variables selected are those with non-zero coefficients.

(b) The penalty $\lambda \sum_j |\beta_j|$ treats all coefficients on the same footing, so the variables must be on comparable scales; otherwise the penalty falls unequally on variables with large versus small measurement scales.

(c) λ is chosen by, for example, cross-validation — maximising the cross-validated partial log-likelihood. Standard errors are usually not reported because the lasso estimates are biased towards zero, so their standard errors are not meaningful and would understate the precision of the estimates.

4 Interpreting and using the fitted model (Book §3.9–3.11)

4.1 Adjusted and unadjusted hazard ratios

Type: What's wrong here?

For the prostatic cancer data (Example 1.4), a model containing only the treatment indicator gives a coefficient of -1.978 (hazard ratio 0.14). After adjusting for tumour size and Gleason index, the coefficient is -1.113 (hazard ratio 0.329).

(a) Explain in words what the two hazard ratios estimate, and why they differ.

(b) A researcher reports "treatment reduces the hazard of death by 86%", based on the unadjusted model. What is wrong with this statement?

(c) Which hazard ratio is the better summary of the treatment effect, and why?

Solution.

(a) The unadjusted hazard ratio (0.14) compares the treatment groups ignoring tumour size and Gleason index; the adjusted hazard ratio (0.329) compares two individuals on different treatments who have the same size and index. They differ because the prognostic variables are not equally distributed across the treatment groups, so the unadjusted comparison conflates the treatment effect with those differences.

(b) "86% reduction" overstates the effect: it is based on the unadjusted value, which is confounded by tumour size and grade. Properly adjusted, the reduction is about 67% (hazard ratio 0.33) — still substantial, but less extreme.

(c) The adjusted hazard ratio, because it isolates the treatment effect from the effects of size and index. The aim is to estimate the effect of treatment, not the combined effect of treatment plus any imbalance in prognostic factors.

4.2 Estimating the survivor function

Type: Concept & notation

Builds on: Chapter 2

After fitting a Cox model, the estimated survivor function for an individual is

$$\hat{S}_i(t) = \{\, \hat{S}_0(t)\}^{\exp(\hat{\beta}' x_i)}.$$

(a) What is $\hat{S}_0(t)$, and how is $\hat{S}_i(t)$ obtained from it?
(b) With no covariates in the model, show that this formula reduces to the Kaplan–Meier estimate.
(c) In the hypernephroma example, nephrectomy has coefficient -1.412. If the baseline survivor function at 12 months is $\hat{S}_0(12) = 0.486$, estimate the 12-month survival probability for a patient aged < 60 who had a nephrectomy, and explain why the two estimated curves cannot cross.

Solution.

(a) $\hat{S}_0(t)$ is the estimated baseline survivor function (for an individual with all covariates equal to zero). $\hat{S}_i(t)$ raises it to the power $\exp(\hat{\beta}' x_i)$, the individual's estimated hazard ratio relative to baseline.
(b) With no covariates, $\exp(\hat{\beta}' x_i) = 1$ and $\hat{S}_i(t) = \hat{S}_0(t)$, which equals the Kaplan–Meier estimate $\prod (n_j - d_j)/n_j$. So the Cox-based estimate generalises the Kaplan–Meier estimate to the case with covariates.
(c) $\hat{S}(12) = 0.486^{\exp(-1.412)} = 0.486^{0.244} \approx 0.84$. Because of proportional hazards the two survivor functions are powers of one another, so one is always above the other and they never cross; the nephrectomy curve lies above (nephrectomy improves prognosis).

5 Mini-lab: fitting a Cox model in R

5.1 Fitting a Cox model in R

Type: Mini-lab

Run the code below. It simulates 200 patients with a treatment indicator (`treat`) and age, generates survival times with a proportional-hazards structure, censors them, and fits a Cox model.

```
set.seed(2718)
n <- 200
treat <- rep(c(0, 1), each = n/2)
age <- round(runif(n, 45, 80))
eta <- 0.55 * treat + 0.04 * (age - 60)
t <- rexp(n, rate = 0.08 * exp(eta))
cens <- runif(n, 0, 60)
obs <- pmin(t, cens)
status <- as.numeric(t <= cens)
d <- data.frame(time = obs, status = status, treat = treat, age = age)

library(survival)
fit <- coxph(Surv(time, status) ~ treat + age, data = d)
summary(fit)
```

```
Call:
coxph(formula = Surv(time, status) ~ treat + age, data = d)

  n= 200, number of events= 172

          coef exp(coef) se(coef)     z Pr(>|z|)
treat 0.641981  1.900242 0.162533 3.950 7.82e-05 ***
age   0.037207  1.037908 0.007944 4.683 2.82e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

      exp(coef) exp(-coef) lower .95 upper .95
treat     1.900     0.5262     1.382     2.613
age       1.038     0.9635     1.022     1.054

Concordance= 0.603  (se = 0.026 )
Likelihood ratio test= 29.62  on 2 df,   p=4e-07
Wald test            = 29.56  on 2 df,   p=4e-07
Score (logrank) test = 29.97  on 2 df,   p=3e-07
```

(a) From the summary, report the estimated hazard ratio and its 95% confidence interval for treatment, and interpret.
(b) Report the hazard ratio per additional year of age, and interpret.
(c) How many events are there? What does the "Concordance" line measure, and is the model useful for discrimination?
(d) What are the three tests at the bottom of the output, and do they agree?

Solution.

(a) The hazard ratio for `treat` is 1.90 (95% CI 1.38 to 2.61). The hazard of the event at any time is about 1.9 times as high for treated versus untreated patients, adjusted for age ($P < 0.001$).
(b) The hazard ratio per additional year of age is 1.038 (95% CI 1.022 to 1.054); each extra year increases the hazard by about 3.8%.
(c) There are 172 events. Concordance (the c-statistic) is 0.60, which is modest: the model separates higher- from lower-risk individuals only weakly (0.5 means no discrimination; values above about 0.7 are generally considered reasonable).
(d) The likelihood ratio, Wald and score (log-rank) tests all test the joint hypothesis that both coefficients are zero. They all give $P < 0.0001$, strongly rejecting the null — the three tests agree.